

# WASP-35b

R-1691

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-35\_150120 · created 2026-08-02T13:12:22+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                                  |
|---------------------------------------------------|----------------------------------|
| Mid-Transit Time (Tmid)                           | 2457042.70921 +/- 0.0002 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.1284 +/- 0.0081                |
| Transit depth (Rp/Rs)^2                           | 1.65 +/- 0.21 [%]                |
| Orbital Inclination (inc)                         | 88.96 +/- 0.76                   |
| Airmass coefficient 1 (a1)                        | 1.1889 +/- 0.0062                |
| Airmass coefficient 2 (a2)                        | -0.1127 +/- 0.0069               |
| Scatter in the residuals of the lightcurve fit is | 0.52 %                           |
| Best Comparison Star                              | None                             |
| Optimal Aperture                                  | 4.5                              |
| Optimal Annulus                                   | 6.0                              |
| Transit Duration (day)                            | 0.1346 +/- 0.0024                |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                             |
|---------------------------|---------------------------------------------|
| Rp/R■ fit vs published    | 0.1284 ± 0.0081 vs 0.1241 (deviation 0.53σ) |
| Duration fit vs published | 0.1346 d vs 0.1317 d (deviation 1.22σ)      |
| Mid-time O–C              | -0.4 min                                    |
| Depth SNR                 | 15.9                                        |
| Residual scatter          | 0.52 %                                      |

## Light curve

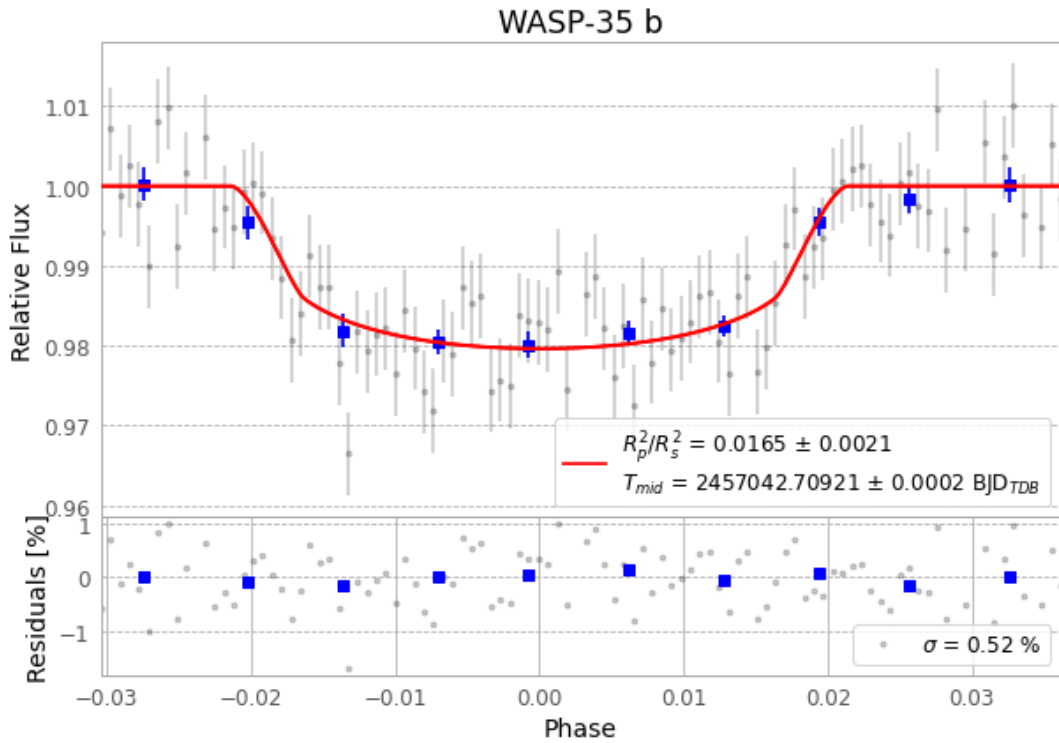


Figure 1 — FinalLightCurve\_WASP-35 b\_2015-01-19.png (SHA-256 b624304c279929a4...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [185, 293], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 0d1d54016dc0c66f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

102 FITS frames (67 MB), WASP-35150120023612.FITS → WASP-35150120073912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-150120.log (f67acf6e005e...), FinalLightCurve\_WASP-35 b\_2015-01-19.png (b624304c2799...), FinalLightCurve\_WASP-35 b\_2015-01-19.csv (407010d4f366...)

## Compute resources

Wall time 175.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 13:12Z.